
Discovering Cooperative Pipelines: Autoresearch for Sequential Social Dilemmas

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Abstract

1 We study *two-level autoresearch* for cooperation: an outer-loop AI agent au-
2 tonomously redesigns the inner-loop pipeline of an LLM policy-synthesis sys-
3 tem for multi-agent Sequential Social Dilemmas (SSDs). A *researcher agent* $\mathcal{R}$
4 (run as a coding agent) reads the inner-loop source code, edits system prompts,
5 feedback functions, helper libraries, and iteration logic, runs evaluations, and
6 decides what to keep, following the *autoresearch* paradigm. Across two games
7 (Cleanup and Gathering), two policy-synthesizer LLMs, and two welfare objec-
8 tives (utilitarian efficiency and Rawlsian maximin), the researcher reliably exceeds
9 hand-designed baselines, sharply tightens run-to-run variance, and outperforms
10 prompt-only optimization. The discovered pipelines are objective-dependent: un-
11 der maximin the researcher independently rediscovers *duty rotation* as a fairness
12 mechanism, supporting an *information-design* reading in which the researcher
13 chooses what to reveal to the boundedly rational synthesizer. Code at <https://github.com/anon590/llm-policies-social-dilemmas>.
14

15 1 Introduction

16 Sequential Social Dilemmas (SSDs) [1] are the multi-agent analogue of the prisoner’s dilemma in
17 temporally rich Markov games: individually rational play leads to collectively suboptimal outcomes
18 through pollution, over-harvesting, or open conflict. Standard multi-agent reinforcement learning
19 (MARL) struggles in this regime due to credit assignment, non-stationarity, and large joint action
20 spaces [2]. A complementary approach, recently introduced by Gallego [3], replaces parameter-
21 space optimization with *algorithm-space synthesis*: a frozen LLM writes a Python policy function,
22 evaluates it in self-play, and iteratively refines it from performance feedback. A single generation
23 step can produce coordination logic (territory partitioning, role assignment, conditional cooperation)
24 at a sample efficiency several orders of magnitude beyond what gradient-based MARL achieves on
25 the same environments.

26 This shifts where the design problem lives, rather than removing it. The inner-loop pipeline that
27 drives the synthesizer has many free parameters: which system prompt, which feedback variables,
28 which helper functions, how many refinement steps. Each materially affects the resulting policies,
29 and prior work tuned them by hand. A natural question follows: *can an AI agent design the pipeline?*

30 We answer affirmatively with a two-level autoresearch framework. An *outer-loop researcher agent*
31 (Claude Opus 4.6, run as a coding agent) edits the source files of an *inner-loop policy synthesizer*
32 (another LLM), runs evaluations on held-out seeds, and keeps modifications that improve a fixed
33 welfare objective Φ . The outer agent operates on an ordinary git repository (reading code, writing
34 diffs, running shell commands, etc) without task-specific scaffolding, mirroring the autoresearch
35 paradigm of Karpathy [4] for single-GPU LLM pretraining. Although the inner-loop SSDs are
36 gridworld benchmarks rather than physical systems, the outer-loop discovery process itself runs

under conditions a deployed discovery agent faces: noisy multi-seed evaluations, stochastic code generation, an LLM-evaluation budget that bounds how often Φ can be queried, and a heterogeneous code repository the agent must navigate end-to-end on its own.

Our contributions are: i) a general two-level framework that delegates the design of an LLM synthesis pipeline to a coding agent operating on a real software repository (Section 3); ii) the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain, with experiments across two SSDs (Cleanup and Gathering), two policy LLMs, and two welfare objectives (utilitarian efficiency U and Rawlsian maximin $\min_i R_i$) (Section 4); and iii) a mechanism-design interpretation supported by the qualitatively different pipelines the agent produces under different welfare objectives, including the autonomous discovery of *time-based duty rotation* as a fairness mechanism, never found under efficiency optimization.

2 Background

We build on the iterative LLM policy synthesis framework of Gallego [3], which serves as the (frozen) inner loop of our two-level system. This section recalls the SSD formalism, the social outcome metrics, and the base synthesis loop.

2.1 Sequential Social Dilemmas

A Sequential Social Dilemma is a partially observable Markov game $\mathcal{G} = \langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, T, \{R_i\}_{i=1}^N, H \rangle$ with N agents, state space $\mathcal{S}$ (the gridworld configuration), per-agent action spaces $\mathcal{A}_i$, transition function T , reward functions R_i , and episode horizon H [1]. Beyond the dilemma’s matrix-game structure, SSDs add temporal richness: agents must learn *when* and *where* to cooperate, not just *whether* to.

We study two canonical SSDs that capture complementary dilemma types.

Cleanup [5] is a *public goods provision* game ($N=10$). The map has two regions: a river that accumulates waste, and an orchard where apples grow. Apples regrow only when river pollution is below threshold. Each agent can fire a cleaning beam (cost -1) that removes waste, collect apples ($+1$ each), or fire a tagging beam (cost -1 , inflicting -50 on the target and removing it for 25 steps). The dilemma: cleaning is costly but its benefits are public, so purely self-interested agents free-ride.

Gathering [1, 6] is a *common pool resource* game ($N=4$). Agents collect shared apples on a fixed respawn timer and may fire tagging beams to temporarily remove rivals. The dilemma: agents can coexist and share resources, or aggress to monopolize them; aggression wastes time and reduces total welfare.

Both games use 8–9 discrete actions (movement, rotation, beam, stand, optionally clean) and episodes of $H=1000$ steps. The two dilemmas differ in cost structure: asymmetric provision (cleaners pay, all benefit) vs. symmetric restraint (every agent faces the same temptation), a distinction that drives our experimental findings.

Social metrics. Following Perolat et al. [6], let $R_i = \sum_{t=0}^{H-1} r_i^t$ denote agent i ’s episode return. We evaluate four social outcomes:

$$U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{efficiency}) \quad (1)$$

$$E = 1 - \frac{\sum_{i,j} |R_i - R_j|}{2N \sum_i R_i} \quad (\text{equality}) \quad (2)$$

$$S = \frac{1}{N} \sum_{i=1}^N \bar{t}_i \quad (\text{sustainability}) \quad (3)$$

$$P = \frac{1}{H} \sum_{t=0}^{H-1} |\{i : \text{active}_i^t\}| \quad (\text{peace}) \quad (4)$$

where $\bar{t}_i$ is the mean timestep at which agent i collects positive reward (higher $\Rightarrow$ resources preserved later) and active_i^t indicates that agent i is not tagged out at step t . We additionally consider the

76 *maximin* (Rawlsian) welfare criterion $\min_i R_i$, which optimizes the worst-off agent’s return and
 77 serves as the second objective in our experiments.

78 2.2 Iterative LLM Policy Synthesis

79 Let Π denote the space of *code-based policies*: deterministic functions $\pi : \mathcal{S} \times [N] \rightarrow \mathcal{A}$ expressed
 80 as executable Python code. Each policy has access to the full environment state and a library of
 81 helpers (BFS pathfinding, beam targeting, coordinate transforms). This state access is a deliberate
 82 design choice: programmatic policies operate in algorithm space rather than the reactive observation-
 83 to-action space of neural policies, which lets a single LLM generation step encode rich coordination
 84 logic.

85 A frozen LLM $\mathcal{M}$ acts as the *policy synthesizer*. Given a system prompt p describing the environ-
 86 ment API and a feedback prompt q_k , it produces a new policy

$$\pi_{k+1} = \mathcal{M}(p, q(\pi_k, \mathcal{F}_k)),$$

87 where π_k is the previous policy (its source code) and $\mathcal{F}_k$ is the evaluation feedback. All N agents
 88 execute the same program π_k in self-play. We stress that this is *symmetric*, not *behaviorally ho-*
 89 *mogeneous*: since π_k takes `agent_id` as an argument, a single shared program can induce distinct
 90 per-agent behaviors (cleaner vs. gatherer assignment, time-rotated duty cycles, partitioned territo-
 91 ries; see the synthesized policies in Appendix C). What is shared is the source code; the sampled
 92 action distributions can differ across agents. Evaluation over a set of random seeds S yields the
 93 mean per-agent return $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$. Each generated
 94 policy passes an AST-based safety check (blocking eval, file I/O, network access) followed by a
 95 short smoke test; failures trigger regeneration (up to R attempts) with the error message appended
 96 to the prompt.

97 **Feedback.** We package the previous policy’s source code together with all available evaluation
 98 signals:

$$\mathcal{F}_k = (\text{code}(\pi_k), \bar{r}_k, \mathbf{m}_k, \mathbf{d}), \quad (5)$$

99 where $\mathbf{d}$ contains natural-language definitions of each social metric. The LLM consumes $\mathcal{F}_k$ to
 100 revise and improve the policy. This is a starting point; the choice of feedback content is a single
 101 design decision within a much larger pipeline configuration space: our two-level framework (Sec-
 102 tion 3) opens the full space to automated search.

103 3 Two-Level Framework

104 We introduce a two-level system where a *researcher agent* $\mathcal{R}$ autonomously discovers configurations
 105 that optimize the output of an inner-loop system. While we instantiate this for multi-agent policy
 106 synthesis, the architecture is general: any pipeline where an LLM generates artifacts, evaluates
 107 them, and iterates can serve as the inner loop. The fundamental insight is that the entire inner-loop
 108 codebase is a designable artifact that a code-based agent can search over. Figure 1 illustrates the
 109 architecture.

110 3.1 Configuration Space

111 Let $\mathcal{C}$ denote the space of *pipeline configurations*. Each configuration $c \in \mathcal{C}$ specifies the full inner-
 112 loop setup:

$$c = (p, \phi, \mathcal{H}, \iota) \quad (6)$$

113 where p is the system prompt, ϕ is the feedback construction function (which metrics and diagnostics
 114 to include, how to frame it, whether to inject adaptive hints and thresholds, etc), $\mathcal{H}$ is the helper func-
 115 tion library (auxiliary functions for pathfinding, getting aggregates of useful environment quantities,
 116 etc.), and ι specifies the iteration logic (number of inner iterations K , sampling strategy). Table 1
 117 provides concrete examples. The validation pipeline is part of the frozen inner-loop infrastructure
 118 rather than a configurable component, the researcher cannot modify it in our experiments to prevent
 119 reward hacking.

120 The hand-designed feedback of [3] corresponds to a single fixed instantiation of ϕ . Our framework
 121 opens the full configuration space to automated search.

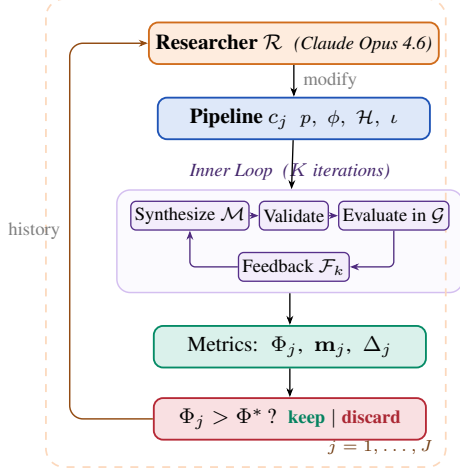


Figure 1: Two-level automated research framework (Algorithm 1).

Algorithm 1 Two-Level Automated Research

Require: Game $\mathcal{G}$, policy synthesizer $\mathcal{M}$, researcher $\mathcal{R}$, system prompt $p_{\mathcal{R}}$, initial config c_0 , outer iterations J , ground-truth objective Φ , held-out seeds S_{ho}

Ensure: Best configuration c^* , best policy π^*

```

1:  $\pi_0^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_0)$ 
2:  $\Phi_0 \leftarrow \text{EVAL}(\pi_0^*; \mathcal{G}, S_{\text{ho}})$ 
3:  $\text{history} \leftarrow \{(c_0, \Phi_0, \mathbf{m}_0, \emptyset)\}$ 
4: for  $j = 1, \dots, J$  do
5:    $c_j \leftarrow \mathcal{R}(p_{\mathcal{R}}, \text{CODE}(c_{j-1}), \text{history})$ 
6:   if  $\neg \text{VALIDATECONFIG}(c_j)$  then  $\text{retry} \leq R$ 
7:      $\pi_j^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_j)$ 
8:      $\Phi_j \leftarrow \text{EVAL}(\pi_j^*; \mathcal{G}, S_{\text{ho}})$ 
9:      $\Delta_j \leftarrow \text{DIFF}(c_{j-1}, c_j)$  // code diff
10:     $\text{history} \leftarrow \text{history} \cup \{(c_j, \Phi_j, \mathbf{m}_j, \Delta_j)\}$ 
11:  end for
12:  $j^* \leftarrow \arg \max_j \Phi_j$ 
13: return  $c_{j^*}, \pi_{j^*}^*$ 

```

Table 1: Configuration components modifiable by the researcher, with concrete edits that $\mathcal{R}$ made in our runs. The environment simulator, ground-truth evaluation, and policy LLM weights are frozen.

	Scope	Concrete edits by $\mathcal{R}$ (see appendix)
p	System prompt	Multi-section strategic briefing with explicit duty-rotation template (agent_id + step//T) % n (App. B.2, Listing 3)
ϕ	Feedback fn	Thresholded diagnostics: “FAIRNESS ALERT” fires when $\min_i R_i < 0$; “DO NOT REGRESS” guard fires when $U \geq 2.5$ (App. B.3, Listing 4)
$\mathcal{H}$	Helper library	BFS-Voronoi territories with respawn-timer awareness; band-based apple zoning by agent_id (App. B.1, Listings 2, 1)
ι	Iteration logic	Per-condition $K \in \{2, 3\}$; $ S $ walked 5→8→12 for variance control; thinking budget 10–32k (App. B.4, Table 6)

3.2 Inner Loop (Policy Synthesis)

Given a configuration c , the inner loop executes K iterations of LLM policy synthesis:

$$\pi_c^* = \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c) \quad (7)$$

Each iteration k proceeds in four stages, following [3]:

- Synthesize.** The policy LLM $\mathcal{M}$ receives the system prompt p , the previous policy’s source code π_{k-1} , and feedback $\mathcal{F}_{k-1}$ constructed by ϕ . It generates a new Python policy function π_k that has access to full environment state and the helper library $\mathcal{H}$.
- Validate.** The generated code undergoes AST-based safety checks (blocking dangerous operations such as file I/O and network access) followed by a short smoke test. Failures trigger re-generation (up to R retries), with the error message appended to the prompt.
- Evaluate.** All N agents execute the same policy π_k in self-play over $|S|$ random seeds (note the policy is conditional on agent_id). The evaluation yields the mean per-agent reward $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$.
- Feedback.** The feedback function ϕ constructs the prompt for the next iteration from $(\pi_k, \bar{r}_k, \mathbf{m}_k)$, packaging the previous policy’s source code together with the scalar reward, the social metrics vector, their natural-language definitions, and any adaptive diagnostics that ϕ injects.

The inner loop output is evaluated on held-out seeds via a fixed ground-truth objective:

$$\Phi(c) = \text{EVAL}(\pi_c^*; \mathcal{G}, S_{\text{held-out}}) \quad (8)$$

139 where Φ maps from social outcomes to a scalar. We consider two alternative objectives to maximize:

$$\Phi_U = U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{utilitarian efficiency}) \quad (9)$$

$$\Phi_{\min} = \min_i R_i \quad (\text{Rawlsian maximin}) \quad (10)$$

140 Φ_U rewards collective throughput and is indifferent to how reward is distributed across agents,
 141 whereas $\Phi_{\min}$ instead optimizes for the worst-off agent, pressuring the researcher toward configura-
 142 tions that distribute the cost of cooperation.

143 3.3 Outer Loop (Automated Research)

144 The researcher agent $\mathcal{R}$ iteratively modifies the pipeline configuration. Following the autoresearch
 145 paradigm [4], $\mathcal{R}$ operates on the inner-loop codebase as a modifiable artifact, proposing changes,
 146 observing outcomes, and refining. The procedure is formalized in Algorithm 1.

147 At each outer iteration j , the researcher $\mathcal{R}$ receives: i) the **full source code** of the current config-
 148 uration c_{j-1} (prompts, feedback construction, helpers, iteration logic); ii) the **experiment history**:
 149 for each prior iteration, the code diff Δ_i , ground-truth score Φ_i , and social metrics vector $\mathbf{m}_i$; iii)
 150 the **environment source code** (read-only), enabling the researcher to reason about game mechanics.
 151 The researcher proposes a new configuration c_j by generating code modifications. Concretely, $\mathcal{R}$
 152 is a coding agent (Claude Code CLI) that operates on a real software repository: it reads and edits
 153 Python source files, runs shell commands, inspects evaluation outputs, and commits changes to a
 154 dedicated git branch, following the same workflow a human researcher would follow.

155 3.4 Connection to Automated Mechanism Design

156 The two-level structure admits a mechanism design interpretation. The researcher $\mathcal{R}$ acts as a *mech-*
 157 *anism designer*: it controls the information structure (what metrics to reveal, how to frame them),
 158 the action space (what helper functions are available), and the incentive structure (how feedback is
 159 presented) under which the policy synthesizer $\mathcal{M}$ operates. The synthesizer acts as the *agent* within
 160 the designed mechanism.

161 This connects to the automated mechanism design literature [7], where a principal designs rules to
 162 induce desired behavior from self-interested agents. In our setting:

- 163 • The **principal** is the researcher $\mathcal{R}$, optimizing the ground-truth objective Φ .
- 164 • The **agent** is the synthesizer $\mathcal{M}$, optimizing per-agent reward as instructed.
- 165 • The **mechanism** is the configuration c : prompts, feedback, helpers, iteration logic.
- 166 • The **outcome** is the social welfare of the resulting multi-agent policy π_c^* .

167 A crucial distinction from classical mechanism design: $\mathcal{M}$ follows instructions but has *bounded*
 168 *rationality* in the sense that its ability to synthesize effective policies depends on the information
 169 and tools provided. The researcher’s task is thus closer to *information design* [8]: choosing what
 170 to reveal to help $\mathcal{M}$ navigate the cooperation–defection tension. Our experiments (Section 4) show
 171 that the researcher designs qualitatively different information structures depending on the welfare
 172 objective Φ , supporting this interpretation empirically.

173 4 Experiments

174 4.1 Setup

175 We conduct 12 autonomous researcher runs across a factorial design. The researcher agent $\mathcal{R}$ is
 176 Claude Opus 4.6, invoked via the Claude Code CLI as a coding agent. Each run operates on a dedi-
 177 cated git branch of a real Python codebase: $\mathcal{R}$ edits source files in `pipeline/`, executes evaluation
 178 scripts, reads metric outputs, and iterates, without human intervention.

179 **Design.** For Cleanup: 2 policy LLMs $\times$ 2 objectives $\times$ 2 replications = 8 runs. For Gathering: 2
 180 policy LLMs $\times$ 1 objective $\times$ 2 replications = 4 runs. Maximin runs are unnecessary for Gathering
 181 because efficiency optimization alone achieves close to perfect equality.

Table 2: Results. Mean / max across the runs per condition. **Bold** marks the best mean per metric within each Policy LLM $\times$ Game sub-block. Baseline: unmodified, hand-designed pipeline from [3]. GEPA: automated prompt optimization [9] with matched compute budget.

Policy LLM	Target	U	E	$\min_i R_i$
<i>Cleanup — Baseline</i>				
Gemini 3.1 Pro	—	1.93/2.70	0.17/0.62	−159/−84
Sonnet 4.6	—	0.86/1.56	−0.02/0.25	−151/−59
<i>Cleanup — GEPA [9]</i>				
Gemini 3.1 Pro	Φ_U	1.34/1.37	−0.10/−0.05	−149/−126
	$\Phi_{\min}$	1.76/2.76	0.90/0.96	143/245
Sonnet 4.6	Φ_U	1.04/1.20	−0.59/−0.21	−164/−126
	$\Phi_{\min}$	−0.63/1.60	0.77/1.00	−147/6
<i>Cleanup — Two-level Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	3.20 /3.25	0.55/0.61	−211/−182
	$\Phi_{\min}$	3.16/3.19	0.98 /0.98	290 /296
Sonnet 4.6	Φ_U	3.12 /3.14	0.66/0.70	−196/−146
	$\Phi_{\min}$	2.57/2.93	0.91 /0.97	179 /200
<i>Gathering — Baseline</i>				
Gemini 3.1 Pro	—	2.04/2.42	0.90/0.98	412/571
Sonnet 4.6	—	0.03/0.03	0.54/0.54	0/0
<i>Gathering — GEPA [9]</i>				
Gemini 3.1 Pro	Φ_U	2.08/2.35	0.94/0.96	436/518
Sonnet 4.6	Φ_U	1.20/1.23	0.63/0.71	86/123
<i>Gathering — Two-level Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	2.49 /2.51	0.98 /0.98	582 /582
Sonnet 4.6	Φ_U	2.52 /2.52	0.96 /0.98	576 /597

Models. Policy synthesizer $\mathcal{M}$: Gemini 3.1 Pro (Google) or Claude Sonnet 4.6 (Anthropic), to recent state-of-the-art LLMs. Both use extended thinking. We additionally test with Gemma 4 26B-A4B-IT (Google), a smaller open-weight model, to probe the framework’s behavior when the policy synthesizer has substantially lower capability (Appendix A.1).

Baselines. The hand-designed feedback configuration from prior work [3] serves as the initial pipeline c_0 for all runs. On Cleanup ($N=10$), this baseline achieves $U=1.93/2.70$ (mean/max) with Gemini and $U=0.86/1.56$ with Sonnet. We additionally compare against GEPA [9], an automated prompt optimization method that iteratively refines the system prompt via LLM reflection. GEPA optimizes only the system prompt p , whereas our researcher modifies the full pipeline $(p, \phi, \mathcal{H}, \iota)$. We give GEPA a matched compute budget: the same number of optimization steps as outer iterations used by our method, so all in all both methods use a comparable number of environment evaluations.

4.2 Results

Table 2 presents the main results, comparing our autoresearch framework against the hand-designed baseline of [3] and GEPA [9].

Finding 1: The researcher reliably improves over hand-designed baselines and outperforms prompt-only optimization. Every run improves substantially regardless of starting point (Figure 2). On Cleanup, autoresearch lifts both LLMs to $U \approx 3.1\text{--}3.2$ from baselines of 1.93 (Gemini) and 0.86 (Sonnet), nearly closing the gap between them; on Gathering, all four runs converge to $U \in [2.47, 2.52]$ from baselines spanning 0.03–2.42. Run-to-run spread is tight (gaps within 0.05 on Cleanup- Φ_U), suggesting the researcher reliably finds the performance ceiling of each policy LLM via the pipeline modifications it discovers (helpers, prompts, and feedback; see appendices B.1–B.3 for a selection of them). At matched environment queries, autoresearch beats GEPA by 2–3 $\times$ on Cleanup (both Φ_U and $\Phi_{\min}$) and 20% on Gathering, with the gap widening for the weaker policy LLM: GEPA-Sonnet under $\Phi_{\min}$ can collapse to a pathological “everyone cleans, nobody

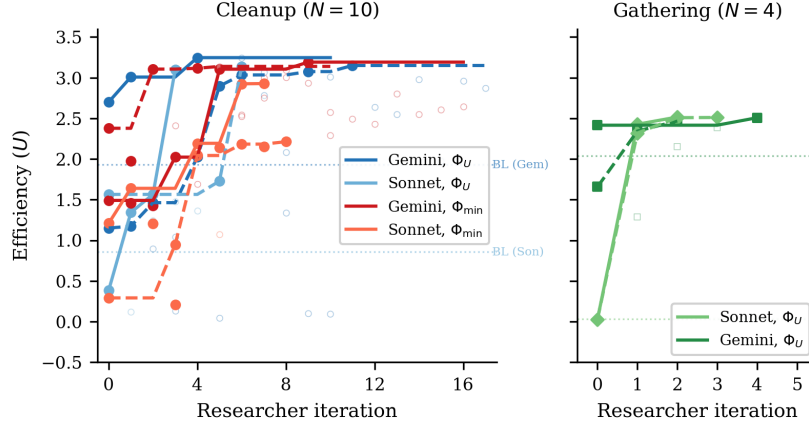


Figure 2: Efficiency (U) across researcher iterations for all 12 runs. *Left*: Cleanup with 8 runs across 2 LLMs $\times$ 2 objectives (solid lines = kept experiments, open circles = discarded). Dashed horizontal line: hand-designed baseline from [3]. All runs converge to $U \approx 3.1$ – 3.2 despite diverse starting points. *Right*: Gathering with 4 runs (2 per LLM). All converge to $U \approx 2.5$ within 2–4 iterations despite baselines ranging from 0.03 to 2.42.

206 eats” regime ($U = -2.87$, $E = 1.00$), while autoresearch-Sonnet reaches $\min_i R_i \approx 200$ reliably.
 207 Modifying the full pipeline, not just the prompt, is what closes these gaps.

208 **Finding 2: No efficiency–fairness tradeoff in Cleanup (Gemini).** Maximin-optimized Gemini
 209 pipelines sacrifice only 1% efficiency (U : 3.16 vs. 3.20) while achieving near-perfect equal-
 210 ity (E : 0.98 vs. 0.55) and transforming maximin from deeply negative baselines (−99 to −84)
 211 to $\min_i R_i = 290$ (Figure 3). The researcher discovers *fair duty rotation* (Listing 6; primed by
 212 the prompt rewrite of Listing 3)—time-based cycling using `agent_id` and `env._step_count`—
 213 simultaneously improving worst-off welfare *and* collective output. Because cleaning is a public
 214 good, distributing the cleaning cost fairly ensures enough cleaners to sustain apple production.

215 For Sonnet, there is a moderate tradeoff: efficiency drops from 3.12 to 2.57 under maximin opti-
 216 mization. The gap reflects Sonnet’s harder time implementing complex coordination mechanisms
 217 (role rotation, zone assignment) from strategic hints alone.

218 **Finding 3: Game structure determines whether fairness requires explicit optimization.** In
 219 Cleanup, where cleaning costs are borne asymmetrically (cleaners pay −1, free-riders collect ap-
 220 ples), baseline equality ranges from $E = 0.04$ to 0.62, and maximin optimization is required to reach
 221 $E > 0.9$. In Gathering, where all agents face a symmetric landscape, efficiency optimization alone
 222 achieves $E > 0.94$ across all 4 runs: no separate maximin runs are needed. This generalizes: in
 223 *provision* dilemmas with asymmetric costs, fairness requires designed mechanisms (role rotation,
 224 duty sharing); in *restraint* dilemmas with symmetric costs, fairness emerges as a free byproduct of
 225 efficient coordination. The researcher independently discovers this, it creates role differentiation
 226 pipelines only for Cleanup, and pure spatial-coordination pipelines for Gathering.

227 **Finding 4: Convergent discovery of qualitatively different strategies per objective.** Despite
 228 fully independent runs, the researcher converges on the same core strategies within each condition
 229 (Table 3, Appendix A). In Cleanup, waste-counting helpers and spatial zone partitioning appear
 230 across all runs. The main qualitative difference is *role rotation*, appearing in 4/4 maximin runs
 231 (Listings 6, 7) but 0/4 efficiency runs (Listing 5). Under efficiency optimization, the researcher
 232 assigns static cleaning roles (some agents always clean), achieving high collective output at the
 233 cost of equality. Under maximin optimization, it discovers rotation as a fairness mechanism. In
 234 Gathering, the researcher discovers BFS-Voronoi territory partitioning and respawn-timer awareness
 235 (Listings 2, 8), with no role differentiation. Thus, optimization is purely spatial (who collects which
 236 apples) and temporal (respawn-aware positioning).

Common failure modes. Several patterns led to discarded autoresearch iterations across all runs: (1) *over-prescription*: adding too many strategic hints confuses the policy LLM, causing generation failures; (2) *iteration regression*: $K=4$ or $K=5$ inner iterations often cause catastrophic regression as the LLM “over-refines” a working policy; (3) *feedback overload*: verbose per-agent diagnostics cause over-correction rather than gradual improvement.

5 Related Work

Sequential social dilemmas. SSDs were introduced by Leibo et al. [1] (Gathering) and extended to public-goods settings by Hughes et al. [5] (Cleanup); Perolat et al. [6] formalized the social outcome metrics (U, E, S, P) used here. Subsequent work studies inequity aversion, intrinsic motivation, and reputational mechanisms for promoting cooperation in MARL agents. We complement this line by automating the search for cooperative *programs* rather than evolving neural policies, and by treating the welfare objective Φ as a designable parameter that the outer agent optimizes for, obtaining qualitatively different cooperative behaviors (static role assignment vs. duty rotation) under different objectives without changing the environment.

LLMs for policy and program synthesis. FunSearch [10] evolves programs for combinatorial discovery; Eureka [11] synthesizes reward functions from environment source code; Voyager [12] and Code as Policies [13] generate executable skill code for single-agent embodied control; ReEvo [14] evolves heuristics with reflective feedback. These works target single-agent settings or non-strategic optimization. Gallego [3], on which our inner loop is based, extended LLM program synthesis to the multi-agent SSD setting, where one program must coordinate N self-play copies. Our contribution is one level above: rather than tuning the synthesizer for one task, we let an outer agent rewrite the synthesis pipeline itself.

LLM reflection and prompt optimization. Reflexion [15], Self-Refine [16], OPRO [17], and GEPA [9] demonstrate that structured verbal feedback loops improve LLM outputs; ERL [18] internalizes such reflection via self-distillation. These methods optimize the prompt or the model’s reasoning trajectory in isolation. Our framework instead modifies the entire surrounding pipeline (system prompt, feedback construction, helper library, iteration logic) making prompt optimization one component of a strictly larger search space. The empirical gap is large: Section 4 shows full-pipeline autoresearch achieving $3\times$ higher efficiency in Cleanup and 37% higher in Gathering than GEPA at matched compute, with run-to-run spread an order of magnitude tighter on the harder fairness objective.

Automated AI research. A small but growing body of work delegates the design of ML pipelines to LLM coding agents. Karpathy’s autoresearch [4] runs a coding agent that modifies `train.py` for nanoGPT pretraining and is rewarded for validation loss. Our system shares this autoresearch architecture (coding agent + frozen evaluation harness + diff-based history) but targets a different inner loop (multi-agent policy synthesis instead of single-model pretraining) and a different objective (multi-agent social welfare instead of validation bits-per-byte). To our knowledge this is the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain.

Automated mechanism and information design. Classical automated mechanism design [7] computes optimal allocation rules for self-interested agents under explicit incentive constraints, while information design [8] studies how a principal commits to a signaling policy that shapes a receiver’s behavior. Our outer agent solves a related but distinct problem: $\mathcal{M}$ is not strategically deceptive (it follows instructions) but is *boundedly rational*, so the researcher must decide what information, helpers, and structure to expose so that $\mathcal{M}$ writes policies achieving the principal’s welfare goal. Section 4 shows that the pipelines $\mathcal{R}$ produces are genuinely a function of the welfare objective, supporting this information-design framing empirically.

6 Discussion and Conclusion

We have presented a two-level framework where a coding agent autonomously discovers pipeline configurations that improve the output of an inner-loop LLM system. Applied to multi-agent policy

synthesis in social dilemmas, the researcher agent reliably exceeds hand-designed baselines across multiple independent runs, and converges on qualitatively similar strategies within each game-objective condition. The framework requires no domain-specific scaffolding: the agent operates on a standard software repository using the same tools (file editing, shell commands, git) available to a human researcher.

Mechanism design in action. Our results empirically support the mechanism design interpretation of Section 3.4. The researcher acts as an information designer: under Φ_U , it reveals efficiency-oriented information (waste counts, zone assignments) that guides $\mathcal{M}$ toward productive but unequal role allocation (Listing 5). Under $\Phi_{\min}$, it additionally reveals fairness-oriented structure such as rotation schedules and per-agent equity feedback (Listings 3, 4), guiding $\mathcal{M}$ toward egalitarian coordination.

Two types of dilemma. The Cleanup–Gathering contrast reveals a structural insight about social dilemmas: *provision* dilemmas (asymmetric costs, one agent’s contribution benefits all) require explicit fairness mechanisms, while *restraint* dilemmas (symmetric costs, each agent faces the same temptation) achieve fairness naturally when coordination improves. This distinction, well-known in the common-pool resource literature, is rediscovered here by an autonomous AI agent, which suggests it is a robust structural feature of these environments.

Human oversight and audit. Our system is fully autonomous by design, but its architecture offers natural affordances for human-in-the-loop oversight. Because the researcher $\mathcal{R}$ operates on a standard git repository, a domain expert can audit the full history of code diffs Δ_j alongside their metric outcomes $(\Phi_j, \mathbf{m}_j)$. The keep/discard decision (Algorithm 1, line 11) is a lightweight intervention point: a human could override the accept threshold, veto modifications that introduce undesirable mechanisms (e.g., exploitative role assignments), or steer the researcher toward under-explored regions of the configuration space by editing the experiment history. More broadly, the separation between the researcher $\mathcal{R}$ and the frozen evaluation Φ creates a *delegation boundary*: the human defines *what* to optimize (the welfare objective), while the agent decides *how*. This mirrors the expert-in-the-loop design patterns identified in deployment-oriented work on AI agents for discovery, where trust calibration depends on the legibility of the agent’s decision trail rather than on real-time supervision.

Future work. Several directions emerge. First, we are interested in applying the two-level framework to other LLM-driven pipelines (code optimization, scientific experiment design, infrastructure tuning), where the same architecture applies with a different inner loop and objective Φ ; next, adversarial objectives can be intriguing, to test whether the researcher discovers exploitative pipeline configurations, exposing reward hacking risks; and asymmetric programs: extend to settings where agents run *different* source code (the present setup is symmetric in code but already supports heterogeneous behavior via `agent_id`; relaxing the shared-program constraint enables richer per-agent specialization and partial-information settings).

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377 A Additional Results

378 **Inner-loop averages confirm a broad-based improvement.** Figures 2 and 3 report the metric of
379 the *kept* inner-loop output (π_c^*) at each outer iteration. A natural concern is that this could overstate
380 pipeline quality if the researcher is benefitting from variance: with K inner iterations per outer
381 step, a single lucky generation could carry the curve while the rest of the inner trajectory is noise.
382 Figures 5 and 6 replot the same runs with the metric averaged across *all* inner iterations of each outer

Table 3: Strategies discovered by the researcher in Cleanup across 4 efficiency-optimized and 4 maximin-optimized independent runs.

Strategy	Φ_U (4 runs)	$\Phi_{\min}$ (4 runs)
Waste-counting helpers	4/4	4/4
Zone/lane partitioning	3/4	4/4
Anti-regression feedback	3/4	2/4
Worked policy examples	2/4	3/4
Cleaning cost economics	2/4	3/4
Time-based role rotation	0/4	4/4

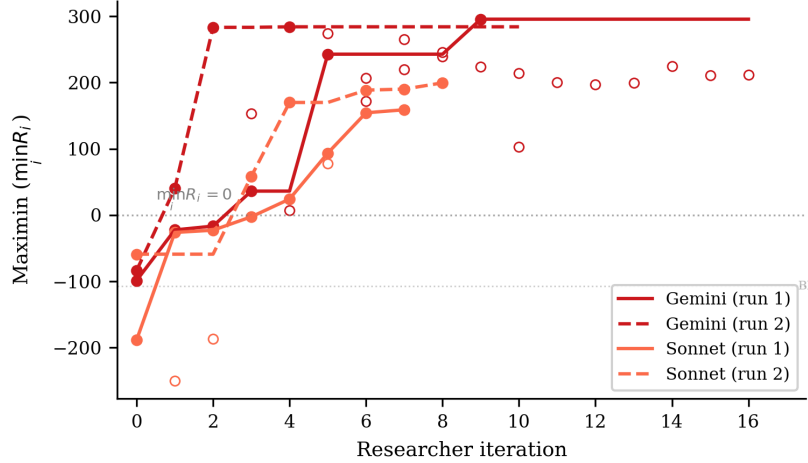


Figure 3: Maximin ($\min_i R_i$) across researcher iterations for the 4 Cleanup maximin-optimized runs. All runs transform deeply negative baselines (worst-off agents losing reward) into substantially positive values. Gemini reaches ~ 290 while Sonnet reaches ~ 160 – 200 . The dashed line marks $\min_i R_i = 0$ (no agent loses reward overall).

step. The trajectories ramp at essentially the same rate, with only mild attenuation: Cleanup mean efficiency still climbs to $\bar{U} \approx 2.5$ – 3.0 (vs. 3.1 – 3.2 for the kept policy), Gathering still saturates at $\bar{U} \approx 2.5$, and Cleanup mean maximin still reaches $\min_i R_i \approx 200$ – 275 (vs. 179 – 290). The whole inner-loop output distribution improves, not just its tail—consistent with the interpretation that the researcher is shaping the synthesizer’s behavior rather than amplifying lucky draws.

A.1 Smaller Open-Weight Model: Gemma 4 26B

We test the framework with Gemma 4 26B-A4B-IT (Google), a 26B-parameter open-weight model substantially smaller than the frontier models in the main experiments. We run three experiment runs: two on Cleanup optimizing efficiency and maximin respectively, and one on Gathering optimizing efficiency. As with the other synthesizer models, all with Opus 4.6 as the researcher $\mathcal{R}$.

Setup. In all three experiments, Gemma starts from complete failure: the baseline pipeline produces $U = -10.0$ on Cleanup (agents CLEAN-spam without collecting apples) and $U = 0.0$ on Gathering (broken BFS calls), compared to $U = 1.93/0.86$ (Gemini/Sonnet on Cleanup) and $U = 2.04/0.03$ (Gemini/Sonnet on Gathering). The researcher ran $J = 10$ – 18 outer iterations per condition.

Results. Table 4 presents the best configurations discovered. Under efficiency optimization, the researcher achieves $U = 0.87$, a dramatic recovery from the broken baseline but far below frontier models ($U \approx 3.2$). The researcher compensates for the model’s limited capability by reducing inner-loop iterations to $K = 1$ (avoiding the regression that plagued $K \geq 2$ runs) and providing highly structured worked examples with explicit cleaning-role assignment.

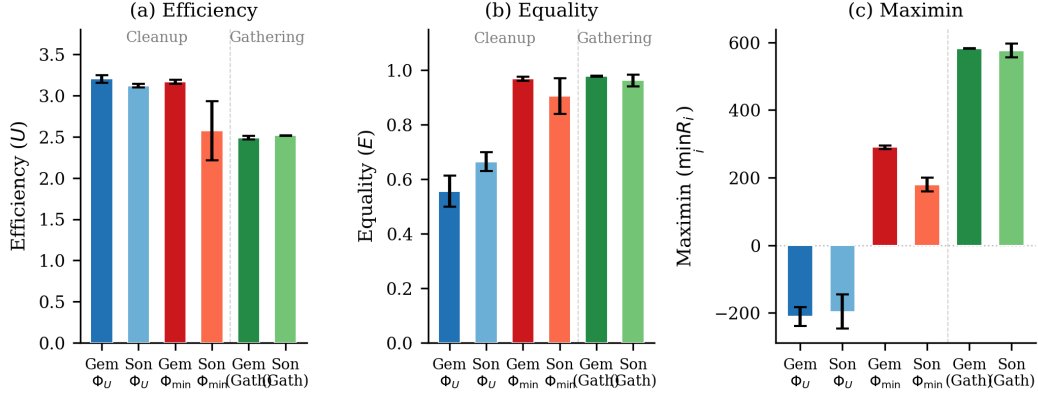


Figure 4: Final metrics across all conditions. (a) Efficiency: all conditions converge to $\bar{U} \approx 2.5$ – 3.2 . (b) Equality: maximin-optimized Cleanup runs achieve $E \approx 1.0$, while efficiency-optimized runs show $E \approx 0.5$ – 0.7 ; Gathering achieves high equality regardless. (c) Maximin: the sharpest contrast—efficiency optimization leaves the worst-off agent deeply negative ($\min_i R_i \approx -200$), while maximin optimization transforms it to $+290$ (Gemini) or $+179$ (Sonnet). Gathering achieves high maximin (~ 580) under efficiency optimization alone. Error bars show s.d. across runs.

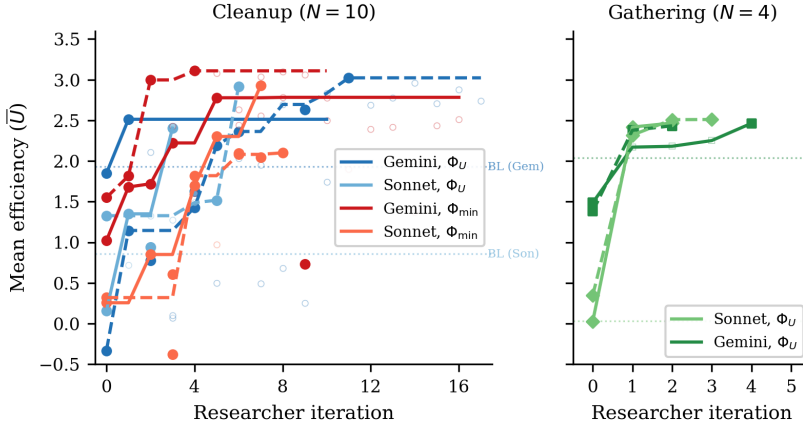


Figure 5: Mean efficiency $\bar{U}$ averaged across all inner iterations of each outer step (compare Figure 2, which shows the kept policy only). *Left*: Cleanup ($N=10$); *right*: Gathering ($N=4$). The trajectories track Figure 2 closely, indicating that the researcher’s gains come from improving the entire inner-loop output distribution, not just the best of K samples.

Maximin optimization rescues efficiency. Strikingly, under maximin optimization the researcher achieves *higher* efficiency ($\bar{U}=1.71$) than under direct efficiency optimization ($\bar{U}=0.87$), alongside strong equality ($E=0.94$) and positive worst-off welfare ($\min_i R_i=137$). This reversal, absent in frontier models where both objectives yield $\bar{U} \approx 3.1$ – 3.2 , occurs because the maximin objective forces the researcher toward coordination mechanisms (rotating cleaning duties, index-based apple assignment) that simultaneously improve collective output. Under efficiency-only optimization, the researcher converges on a local optimum that a 26B model can implement but that caps performance well below the frontier.

This suggests that for models below a capability threshold, fairness objectives may serve as better optimization targets for overall social welfare than direct efficiency maximization. The structured coordination enforced by the maximin objective provides a scaffold that compensates for the weaker model’s difficulty implementing complex strategies from strategic hints alone.

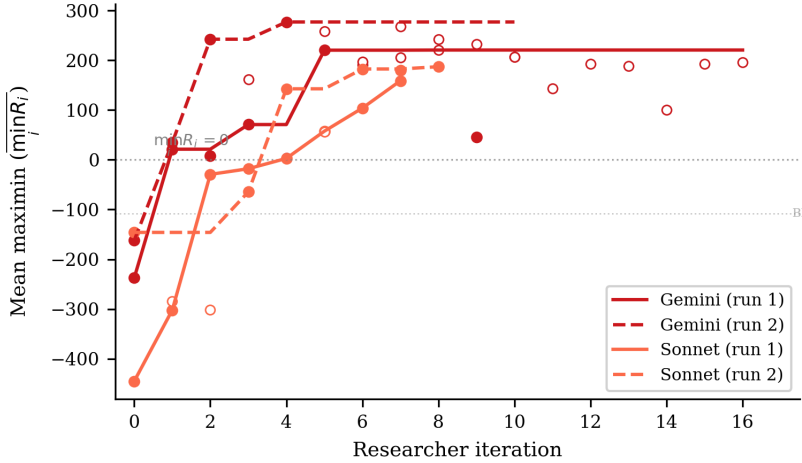


Figure 6: Mean maximin $\overline{\min_i R_i}$ averaged across all inner iterations of each outer step for the 4 Cleanup $\Phi_{\min}$ runs (compare Figure 3). The deeply negative baselines lift to ~ 200 – 275 mean maximin, only mildly below the kept-policy values of Figure 3. The horizontal dashed line marks $\min_i R_i = 0$.

Table 4: Autoresearch with Gemma 4 26B-A4B-IT. Single run per condition. Baseline: pipeline from [3].

Game	Target	U	E	$\min_i R_i$	Keep rate
Cleanup	Baseline	-10.0	1.00 [†]	-1000	—
	Φ_U	0.87	-1.11	-371	6/19
	$\Phi_{\min}$	1.71	0.94	137	9/17
Gathering	Baseline	0.0	1.00 [†]	0	—
	Φ_U	2.44	0.98	580	4/11

[†]Equality is trivially 1.0 because all agents receive near-zero reward.

Gathering: full recovery on a simpler game. On Gathering ($N=4$), the researcher brings Gemma from $U=0.0$ to $U=2.44$ with $E=0.98$ and $\min_i R_i=580$, after 10 outer iterations (4 kept). These values nearly match frontier models (Gemini: $U=2.49$, Sonnet: $U=2.52$; Table 2). The researcher discovers the same Voronoi partitioning and respawn-aware camping strategies found in the main experiments, and increases inner-loop iterations to $K=5$ to allow sufficient refinement. The contrast with Cleanup suggests that the researcher can fully compensate for model weakness on simpler coordination tasks (4 agents, symmetric costs) but not on harder provision dilemmas (10 agents, asymmetric costs).

A.2 Compute Requirements

Table 5 reports wall-clock time for all 12 autoresearch runs. *Inner loop* measures total time spent on policy LLM generation and environment simulation across all evaluations; *total* (estimated from run-directory timestamps) additionally includes the researcher agent’s analysis, code editing, and decision-making between evaluations.

Cost breakdown. Each inner loop evaluation involves $K=2$ – 3 policy LLM generation calls (each ~ 2 k input tokens, ~ 1.5 k output tokens) followed by multi-seed simulation (5–12 seeds, ~ 15 – 30 s total). Policy LLM generation dominates inner loop time (86–97%), with Sonnet evaluations taking $\sim 2\times$ longer than Gemini due to extended thinking. The researcher agent (Claude Opus 4.6, running via Claude Code CLI) accounts for 41% of total wall-clock time (25.6h of the 62.2h total across all 12 runs), spent reading results, editing pipeline source files, and planning modifications. In monetary terms, the researcher agent is the dominant cost: each outer iteration consumes ~ 50 – 100 k

Table 5: Wall-clock time per autonomous run. Evals: total inner loop evaluations including initial baseline. Per eval: mean wall-clock per single evaluation. Total: end-to-end including researcher overhead.

Policy LLM	Φ	Evals	Inner loop (h)	Per eval (min)	Total (h)
<i>Cleanup (N=10)</i>					
Gemini	Φ_U	11	2.6	14	3.7
Gemini	Φ_U	18	4.1	14	6.4
Sonnet	Φ_U	4	2.1	32	6.1
Sonnet	Φ_U	7	5.0	43	6.1
Gemini	$\Phi_{\min}$	17	3.0	11	4.3
Gemini	$\Phi_{\min}$	11	3.6	20	5.0
Sonnet	$\Phi_{\min}$	8	4.5	33	8.8
Sonnet	$\Phi_{\min}$	9	5.3	36	13.2
<i>Gathering (N=4)</i>					
Sonnet	Φ_U	3	1.7	33	2.2
Gemini	Φ_U	5	1.4	17	1.9
Gemini	Φ_U	3	0.9	19	1.1
Sonnet	Φ_U	4	2.3	35	3.4
All 12 runs		100	36.6	22	62.2

context tokens in the Opus session, whereas each inner loop evaluation uses only ~ 5 – 10 k policy LLM tokens.

B Researcher-Authored Pipeline Artifacts

The previous appendix shows the policies π_c^* that the inner loop *outputs*; this one shows the configuration $c = (p, \phi, \mathcal{H}, \iota)$ (Section 4, Table 1) that the researcher $\mathcal{R}$ *authored* to produce them. Excerpts are taken from the final commit on the dedicated git branch of the corresponding run; we reproduce the prose verbatim, with author commentary in [brackets] and elisions marked We organize by artifact type so each subsection makes a within-type contrast (e.g., Φ_U vs. $\Phi_{\min}$, or Cleanup vs. Gathering).

B.1 Helper library $\mathcal{H}$: coordination primitives

The researcher adds primitives that the policy LLM can call as black boxes, sparing it from re-implementing tricky logic on every iteration. Two patterns dominate: (i) *state inspection* (waste counts, fractions, beam-yield scoring) and (ii) *coordination primitives* (zone assignment, role rotation). We show one of each.

Listing 1: Cleanup, $\Phi_{\min}$ – band-based apple zoning helper added by the researcher in exp5. This is the primitive that lets the policy in Listing 6 keep collectors within their own row band, preventing all 7 gatherers from racing to the same apple.

```

448 1 def get_my_apples(env, agent_id):
449 2     """Alive apples in this agent's horizontal row band.
450 3     Divides the orchard into n_agents bands; falls back to all apples if empty."""
451 4     aid = int(agent_id)
452 5     n = int(env.n_agents)
453 6     band_h = env.height / n
454 7     band_lo, band_hi = aid * band_h, (aid + 1) * band_h
455 8
456 9     my_apples, all_apples = set(), set()
457 10    for i in range(env.n_apples):
458 11        if env.apple_alive[i]:
459 12            ar = int(env._apple_pos[i, 0]); ac = int(env._apple_pos[i, 1])
460 13            all_apples.add((ar, ac))
461 14            if band_lo <= ar < band_hi:
462 15                my_apples.add((ar, ac))
463 16    return my_apples if my_apples else all_apples
464

```

Listing 2: Gathering – BFS-Voronoi territory + respawn-aware waiting helpers added by the researcher in gather-exp3. These are the primitives that Listing 8 composes into a one-line policy: “go to my_zone_apples, otherwise nearest_respawning_apple.”

```

466 1 def voronoi_zones(env):
467 2     """Multi-source BFS Voronoi over walkable cells.
468 3     Returns (row, col) -> agent_id; ties broken by lower agent_id."""
469 4     queue, visited = deque(), {}
470 5     for a_id in range(env.n_agents):
471 6         if int(env.agent_timeout[a_id]) == 0:
472 7             ar = int(env.agent_pos[a_id][0]); ac = int(env.agent_pos[a_id][1])
473 8             queue.append((ar, ac, a_id))
474 9             visited[(ar, ac)] = a_id
475 10    while queue:
476 11        r, c, a_id = queue.popleft()
477 12        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
478 13            nr, nc = r + dr, c + dc
479 14            if 0 <= nr < env.height and 0 <= nc < env.width \
480 15                and not env.walls[nr, nc] and (nr, nc) not in visited:
481 16                visited[(nr, nc)] = a_id
482 17                queue.append((nr, nc, a_id))
483 18    return visited
484 19
485 20 def nearest_respawning_apple(env, agent_id, zones=None, max_timer=10):
486 21     """Nearest dead apple in MY zone respawning within max_timer steps.
487 22     Returns (row, col) of best wait spot, or None."""
488 23     if zones is None: zones = voronoi_zones(env)
489 24     best_pos, best_t, best_d = None, max_timer + 1, float('inf')
490 25     ar = int(env.agent_pos[agent_id][0]); ac = int(env.agent_pos[agent_id][1])
491 26     for i in range(env.n_apples):
492 27         if not env.apple_alive[i]:
493 28             pos = (int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
494 29             if zones.get(pos) == agent_id:
495 30                 t = int(env.apple_timer[i])
496 31                 if t <= max_timer:
497 32                     d = abs(pos[0] - ar) + abs(pos[1] - ac)
498 33                     if t < best_t or (t == best_t and d < best_d):
499 34                         best_pos, best_t, best_d = pos, t, d
500 35    return best_pos
501 36
502 37

```

503 The choice of helper depends on the dilemma’s geometry: row-bands suffice when fairness is the
504 main concern (Cleanup maximin) but full BFS-Voronoi is needed when wall-aware territory owner-
505 ship matters (Gathering). The researcher does not fix this in advance, it picks the simpler primitive
506 whenever it works.

507 B.2 System prompt p : from neutral framing to strategic briefing

508 The unmodified system prompt is a neutral API description: “Write a policy that maximizes per-
509 agent reward.” Under maximin optimization, the researcher rewrites the prompt into a multi-section
510 strategic briefing. The excerpt below shows the pieces it added to the Cleanup prompt in exp5; the
511 full prompt grew from 165 to 325 lines.

Listing 3: Cleanup, $\Phi_{\min}$ – excerpts from the researcher-rewritten system prompt (exp5). The unmodified baseline contained only API documentation; everything below was added by $\mathcal{R}$.

```

512 ## CRITICAL OBJECTIVE: Rawlsian Fairness (Maximin)
513
514 Your goal is to maximize the minimum per-agent total return across all
515 agents. This is the "maximin" or Rawlsian welfare criterion.
516
517 This means: it is NOT enough for the *average* agent to do well. The
518 worst-off agent must do as well as possible. A policy where 8 agents
519 each earn +200 but 2 agents each earn -100 is TERRIBLE (maximin = -100).
520
521 Key implication: cleaning costs (-1 per CLEAN action) must be shared
522 equitably among ALL agents. If you assign fixed "cleaner" roles, those
523 agents will accumulate large negative rewards and destroy your maximin score.
524
525 ## Strategy for Maximin: ROLE ROTATION + APPLE ZONING
526
527 The optimal strategy for maximin involves:
528 1. Shared cleaning duty: ALL agents take turns cleaning using a rotation
529    schedule based on 'agent_id' and 'env._step_count'. For example:
530    'is_my_cleaning_turn = (agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS'
531    where SHIFT is ~50 steps and NUM_CLEANERS is 2-3.
532 2. When NOT your turn: collect apples from YOUR ZONE using
533

```

```

534     'get_my_apples(env, agent_id)' -- prevents all gatherers competing
535     for the same nearest apple.
536 3. NEVER use BEAM (action 6): it costs -1 and causes -50 to the target.
537 4. Keep waste density low: aim for ~2-3 active cleaners at any time.
538
539 [... API documentation, helper documentation, working example ...]
540
541 IMPORTANT:
542 - NEVER assign permanent cleaning roles to specific agents -- this kills maximin.
543 - Use env._step_count for time-based role rotation so all agents share cleaning duty.
544 - NEVER use BEAM (action 6) -- it destroys both agents' rewards.

```

Two things stand out. First, the researcher writes the formula it expects $\mathcal{M}$ to use almost verbatim (`(agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS`) and the policy in Listing 6 uses essentially this template. Second, the researcher *teaches by counter-example*: the explicit “8 agents earn +200, 2 earn -100, maximin = -100” makes the failure mode of Φ_U -style static roles (Listing 5) concretely visible to $\mathcal{M}$. This is the information-design move predicted by the mechanism-design framing in Section 3.4.

The Gathering prompt evolves along a different axis (no maximin, no rotation). Its researcher-added “Key Strategic Insights” section instead emphasizes (i) *self-play implies never beam* and (ii) *Voronoi partition each step*, exactly matching what the policy in Listing 8 implements.

555 B.3 Feedback ϕ : adaptive diagnostics

The baseline feedback function from [3] shows reward + four social metrics with definitions and stops there. The researcher turns it into a state machine: it inspects the latest metrics and conditionally injects different hints. Two different runs produced two qualitatively different diagnostics: the same metric ($\bar{r}_k$ or $\min_i R_i$) is repurposed to either *stabilize* a working policy or *redirect* a failing one.

Listing 4: Adaptive feedback diagnostics. *Top*: efficiency-optimized run (exp1) – a stability guard prevents iter-3 regression once U is high. *Bottom*: maximin-optimized run (exp5) – two fairness diagnostics that fire when the worst-off agent loses reward.

```

561 1 # --- Cleanup, Phi_U feedback (exp1, pipeline/feedback.py final) ---
562 2 last_eff = history[-1]["metrics"]["efficiency"]
563 3 if last_eff >= 2.5:
564 4     parts.append(
565 5         """**CRITICAL -- DO NOT REGRESS**: The current policy achieves high "
566 6         "efficiency (>=2.5). You MUST output a policy that is nearly identical "
567 7         "to the current one. Copy the current policy code and make AT MOST "
568 8         "one small targeted improvement (e.g., adjust a single numeric "
569 9         "threshold). If you are not confident a change will help, output the "
570 10        "current policy UNCHANGED. A regression here means the run fails.")
571 11
572 12 # --- Cleanup, Phi_min feedback (exp5, pipeline/feedback.py final) ---
573 13 last_maximin = history[-1]["metrics"].get("maximin", 0)
574 14 last_avg = history[-1]["reward_avg"]
575 15 if last_maximin < 0:
576 16     parts.append(
577 17         f"""**FAIRNESS ALERT**: maximin={last_maximin:.1f} is NEGATIVE. "
578 18         f"The worst-off agent lost reward overall while average was {last_avg:.1f}. "
579 19         "This means cleaning duties are NOT shared equitably. "
580 20         "Use time-based role rotation (env._step_count) so ALL agents share "
581 21         "cleaning costs equally. NEVER assign permanent cleaner roles.")
582 22 elif last_maximin < last_avg * 0.5:
583 23     parts.append(
584 24         f"""**FAIRNESS WARNING**: maximin={last_maximin:.1f} is much lower than "
585 25         f"average={last_avg:.1f}. The gap suggests unequal cleaning burden. "
586 26         "Ensure ALL agents rotate through cleaning duty.")
587 26

```

The two diagnostics are written by independent researcher runs but converge on a common pattern: *thresholded interventions on a primary metric*. They are also a common cause of the low run-to-run variance reported in Finding 1: when a working policy lands in the high- U basin, the stability guard prevents the LLM from over-refining and tipping out of it (the $K=3$ regressions visible in Section 4’s “Common failure modes” (4)), and when an unfair policy lands in the negative-maximin region, the alert injects exactly the rotation hint that recovers it.

595 B.4 Iteration logic ι : per-condition hyperparameters

596 Although ι has the smallest source surface, the researcher’s edits to it explain a meaningful fraction
 597 of the variance reduction noted in Finding 1. Table 6 summarizes the values shipped in the final
 598 commit of each best-performing run.

Table 6: Iteration-logic (ι) values in the final commit of selected runs. K = inner-loop iterations, $|S|$ = evaluation seeds, “thinking” = extended-thinking token budget passed to $\mathcal{M}$. Baseline ([3]) is $K=3$, $|S|=5$, thinking 16k.

Run	Game / objective	K	$ S $	thinking	Researcher’s stated rationale
exp1 (Gemini)	Cleanup, Φ_U	3	5	16k	default; stability comes from feedback hint
exp4 (Sonnet)	Cleanup, Φ_U	3	5	32k	“Sonnet was over-refining at 16k thinking”
exp5 (Gemini)	Cleanup, $\Phi_{\min}$	2	12	16k	“ $K=2$ avoids iter-3 regression; 12 seeds for variance control”
exp7 (Sonnet)	Cleanup, $\Phi_{\min}$	3	5	10k	“reduced thinking from 16k – Sonnet was generating runaway code at 16k”
gather-exp3 (Gemini)	Gathering, Φ_U	3	5	16k	default; Gathering converges in $K \leq 3$

599 The maximin Gemini run is the most informative case: the researcher discovered that with $K=3$
 600 and $|S|=5$, identical configurations could yield $\min_i R_i=295$ on one set of seeds and $\min_i R_i=100$
 601 on another (this is the “variance exposure” failure mode of Section 4’s common failures). It then
 602 walked $|S|$ from $5 \rightarrow 8 \rightarrow 12$ over three outer iterations until the seed-averaged signal was stable
 603 enough that the policy LLM stopped chasing noise. The policy in Listing 6 is sampled at $|S|=12$.

604 B.5 Cross-artifact observations

- 605 • *Helpers do the algebra; prompts pick the strategy class.* The researcher uses helpers (1, 2) to
 606 put non-trivial primitives at $\mathcal{M}$ ’s fingertips, and the prompt (3) to tell $\mathcal{M}$ *which* primitive to
 607 compose. Prompts without supporting helpers produced policies that “mention rotation” but failed
 608 to implement it; helpers without prompt updates often went unused.
- 609 • *Feedback is the only adaptive component.* p , $\mathcal{H}$, and ι are static within a run; ϕ is the only place
 610 where the researcher gets to change what $\mathcal{M}$ sees *during* the inner loop, and it uses thresholded
 611 diagnostics (Listing 4) to do so. This is what mostly drives the run-to-run variance reduction.
- 612 • *The same artifact has different content under each objective.* The Cleanup prompt under Φ_U omits
 613 “rotation” and “maximin” entirely; the same prompt under $\Phi_{\min}$ devotes its entire opening section
 614 to it. The configuration c is genuinely a function of the welfare objective Φ , not a fixed pipeline
 615 parameterized by it. This is consistent with the information-design framing of Section 3.4.
- 616 • *The researcher’s edits are short and targeted.* The full diff between the baseline pipeline and the
 617 best maximin Gemini configuration totals ~ 300 added lines spread over four files; ~ 80 of them
 618 appear in the listings above. The remaining content is API documentation and worked examples,
 619 included for completeness but not strategically novel.

620 C Selected Generated Policies

621 The code excerpts below are verbatim outputs of the policy synthesizer $\mathcal{M}$, taken from the final
 622 inner-loop iteration of the best-performing run in each condition. To fit the page, we elide repeated
 623 boilerplate (BFS scaffolding, fallback branches) with $\dots$ and add commentary in [brackets];
 624 nothing else has been edited. The listings illustrate the qualitative differences highlighted in Findings
 625 2–4 of Section 4: *static* vs. *rotating* cleaning roles in Cleanup, and the *Voronoi* + *respawn-timer*
 626 structure in Gathering. All four policies were independently generated by the LLM under different
 627 seeds, models, and outer-loop runs.

628 C.1 Cleanup, Φ_U (Gemini): static interleaved roles + dynamic threshold

629 Best run: $U=3.25$, $E=0.61$, $\min_i R_i=-182$. The synthesizer assigns *permanent* cleaning roles via
 630 $\text{agent_id} \bmod n < \tau$, where the threshold τ depends smoothly on the current waste fraction. Higher
 631 pollution recruits more cleaners, but the *identity* of each cleaner is fixed: agents $0-\tau-1$ always pay
 632 the cost. This drives U to its ceiling but leaves equality at 0.61 and pushes the worst-off agent to
 633 $\min_i R_i=-182$.

Listing 5: Cleanup, Φ_U – Gemini 3.1 Pro. Static role assignment with waste-tiered threshold (verbatim, abridged).

```

634 1 def policy(env, agent_id) -> int:
635 2     if int(env.agent_timeout[agent_id]) > 0:
636 3         return 7 # STAND while removed
637 3
638 4
639 5     waste_frac = waste_fraction(env)
640 6     n_agents = env.n_agents
641 7
642 8     # [Smooth dynamic threshold: more pollution -> more cleaners]
643 9     if waste_frac > 0.35: threshold_cleaners = n_agents
644 10    elif waste_frac > 0.25: threshold_cleaners = int(n_agents * 0.7)
645 11    elif waste_frac > 0.15: threshold_cleaners = int(n_agents * 0.5)
646 12    elif waste_frac > 0.05: threshold_cleaners = int(n_agents * 0.3)
647 13    else: threshold_cleaners = max(1, int(n_agents * 0.2))
648 14
649 15    # [STATIC role: cleaners are ALWAYS the lowest-id agents -- no rotation]
650 16    is_cleaner = (agent_id % n_agents) < threshold_cleaners
651 17
652 18    if is_cleaner:
653 19        best_orient, n_waste = find_best_clean_orientation(env, agent_id)
654 20        if n_waste >= 1:
655 21            cur_orient = int(env.agent_orient[agent_id])
656 22            if best_orient == cur_orient: return 8 # CLEAN
657 23            if (cur_orient + 1) % 4 == best_orient: return 5 # ROTATE_RIGHT
658 24            elif (cur_orient - 1) % 4 == best_orient: return 4 # ROTATE_LEFT
659 25            else: return 4
660 26            # [Competitor-aware BFS toward waste in own row band]
661 27            c_idx = agent_id % n_agents
662 28            target_r = int(((c_idx + 0.5) / threshold_cleaners) * env.height)
663 29            # ... pick waste cell minimizing dist + |row - target_r| + 50 *
664 30            # closer_cleaners
665 31            # ... bfs_toward and direction_to_action
666 32        else:
667 33            # GATHERER: row-banded sweep, competitor-aware
668 34            # assigned_row = ((g_idx + 0.5) / n_gatherers) * env.height
669 35            # ... pick alive apple minimizing dist + |row - assigned_row| + 50 *
670 36            # closer_gatherers
671 37            ...

```

673 C.2 Cleanup, $\Phi_{\min}$ (Gemini): time-rotated, geographically interleaved roles

674 Best run: $U=3.19$, $E=0.98$, $\min_i R_i=296$. Switching the outer objective from U to $\min_i R_i$ flips
675 the role-assignment scheme entirely: `role_idx` now depends on `agent_id + step/50`, so every
676 agent rotates through cleaner and gatherer duty in a 50-step cycle. The cleaner indices $\{1, 4, 8\}$ are
677 spatially *interleaved* along the river (rather than contiguous) so that whoever is on duty has a short
678 walk to their assigned river slice. This single change accounts for the $E=0.55 \rightarrow 0.98$ jump and the
679 $\min_i R_i=-182 \rightarrow 296$ swing reported in Table 2, with negligible efficiency loss.

Listing 6: Cleanup, $\Phi_{\min}$ – Gemini 3.1 Pro. Time-rotated roles with interleaved geography (verbatim, abridged).

```

680 1 def policy(env, agent_id) -> int:
681 2     if int(env.agent_timeout[agent_id]) > 0:
682 3         return 7 # STAND while removed
683 3
684 4
685 5     aid = int(agent_id)
686 6     step = int(env._step_count)
687 7     n = int(env.n_agents)
688 8
689 9     # [TIME ROTATION: every 50 steps each agent's role index advances]
690 10    if n == 10:
691 11        role_idx = (aid + step // 50) % 10
692 12        # [Cleaner indices {1,4,8} are SPATIALLY INTERLEAVED, not contiguous,
693 13        # so each on-duty cleaner has a short walk to its river slice]
694 14        if role_idx == 1: role_type, zone_idx = 'C', 0
695 15        elif role_idx == 4: role_type, zone_idx = 'C', 1
696 16        elif role_idx == 8: role_type, zone_idx = 'C', 2
697 17        else:
698 18            role_type = 'G'
699 19            g_map = {0:0, 2:1, 3:2, 5:3, 6:4, 7:5, 9:6}
700 20            zone_idx = g_map[role_idx]
701 21            num_cleaners, num_gatherers = 3, 7
702 22
703 23    if role_type == 'C':
704 24        # [Each on-duty cleaner owns one horizontal slice of the river]

```

```

705 25     row_min = int( zone_idx      * env.height / num_cleaners)
706 26     row_max = int((zone_idx + 1) * env.height / num_cleaners) - 1
707 27
708 28     # [Score every (row, col, orientation) in the slice by beam yield;
709 29     # fire if facing >=2 waste, else move to a >=2-yield position]
710 30     best_positions, good_positions = set(), set()
711 31     for r in range(row_min, row_max + 1):
712 32         for c in range(10):
713 33             if env.walls[r, c] or env.waste[r, c]: continue
714 34             y_max = max(get_clean_yield(env, r, c, o) for o in range(4))
715 35             if y_max >= 2: best_positions.add((r, c))
716 36             if y_max >= 1: good_positions.add((r, c))
717 37         # ... rotate / CLEAN / bfs_to_target_set as appropriate
718 38     else:
719 39         # GATHERER: collect apples ONLY inside the rotating row band
720 40         # row_min/row_max for num_gatherers slices ...
721 41     ...

```

723 C.3 Cleanup, $\Phi_{\min}$ (Sonnet): independent rediscovery of duty rotation

724 Best run: $U=2.93$, $E=0.83$, $\min_i R_i=154$. Run with a different policy LLM (Sonnet 4.6) and on
725 a different outer-loop trajectory, the synthesizer arrives at the *same* structural insight as the Gemini
726 maximin run: a phase counter ($\text{agent_id} + \text{step}/50$) mod n rotates which 2 of the 10 agents clean
727 at any given time, and a *separate* 200-step zone counter rotates which 5-row band each collector
728 sweeps. The two rotation periods (50 and 200 steps) ensure every agent visits every cleaning slot
729 and every apple zone within one episode, resulting in a structural fairness invariant the researcher
730 discovered without any explicit instruction to “rotate.”

Listing 7: Cleanup, $\Phi_{\min}$ – Sonnet 4.6. Phase-rotated cleaning + zone-rotated collection (verbatim, abridged).

```

731 1 def policy(env, agent_id) -> int:
732 2     if int(env.agent_timeout[agent_id]) > 0:
733 3         return 7 # STAND while removed
734 4
735 5
736 6     n = env.n_agents
737 7     step = env._step_count
738 8     wf = waste_fraction(env)
739 9
740 10     # --- ROLE ASSIGNMENT: fair 2-cleaner rotation ---
741 11     # [50-step phases x 20 phases / episode -> each agent cleans 4 phases (~20%)]
742 12     phase = step // 50
743 13     cleaner_rank = (agent_id + phase) % n # [cycles 0..9 fairly]
744 14     is_cleaner = cleaner_rank < 2 # [exactly 2 cleaners per phase]
745 15
746 16     # [Emergency override: all agents clean when waste crosses the spawn cliff]
747 17     in_emergency = wf >= 0.40
748 18     if in_emergency: is_cleaner = True
749 19
750 20     # [River split: cleaner_slot 0 -> top half, 1 -> bottom half]
751 21     cleaner_slot = (agent_id % 2) if in_emergency else cleaner_rank
752 22
753 23     if is_cleaner and wf > 0.04:
754 24         # [Scan 4 orientations from current position; if best beam shot
755 25         # covers >=3 waste cells, fire; else step to a strictly better neighbor]
756 26         # ... (best_o, best_cnt) = max over 4 directions
757 27         # ... if best_cnt >= 3: rotate to best_o, then CLEAN
758 28         # ... else: scan 4 adjacent cells x 4 orientations for higher yield
759 29         # ... if no waste in beam range: bfs to waste_set in DOWN river half,
760 30         # fall back to whole river
761 31
762 32     # --- COLLECTOR LOGIC: 5-zone rotating apple collection ---
763 33     # [5 zones x 200 steps = 1000 steps -> each agent visits every zone once;
764 34     # 2 agents share each zone at any time, reducing competition 10 -> 2]
765 35     bounds = zone_boundaries(env, 5)
766 36     zone_phase = step // 200
767 37     zone = (agent_id + zone_phase) % 5
768 38     z_start, z_end = bounds[zone], bounds[zone + 1]
769 39
770 40     zone_apples = {(int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
771 41                    for i in range(env.n_apples) if env.apple_alive[i]
772 42                    and z_start <= int(env._apple_pos[i][0]) < z_end}
773 43     # ... bfs_to_target_set(zone_apples), else bfs_nearest_apple as fallback
774 44     return 7

```

776 C.4 Gathering (Gemini): wall-aware Voronoi + spatiotemporal targeting

777 Best run: $U=2.47$, $E=0.98$, $\min_i R_i=580$. In Gathering, where costs are symmetric, no role differentiation is needed: the entire optimization is spatial and temporal. The synthesizer (i) partitions cells into BFS-Voronoi territories owned by individual agents, (ii) runs a single centralized BFS from its own position to score every reachable cell with exact wall-aware distances, and (iii) for both alive and respawning apples in its territory, computes the *earliest collection time* $\max(\text{walk_dist}, \text{respawn_timer})$ and targets the minimum. When the agent must wait on a dead apple, it steps onto a *non-spawn* adjacent cell rather than blocking a respawn point. This single sort key ($\max(\text{distance}, \text{timer})$) subsumes both “go to nearest apple” and “camp respawn” as special cases.

Listing 8: Gathering – Gemini 3.1 Pro. Voronoi territories + spatiotemporal priority targeting (verbatim, abridged).

```

786 1 def policy(env, agent_id) -> int:
787 2     if int(env.agent_timeout[agent_id]) > 0:
788 3         return 7
789 4
790 5
791 6     r, c = int(env.agent_pos[agent_id][0]), int(env.agent_pos[agent_id][1])
792 7     orient = int(env.agent_orient[agent_id])
793 8
794 9     # [Wall-aware Voronoi over the gridworld; spawn_points = all apple cells]
795 10    zones = voronoi_zones(env)
796 11    spawn_points = {(int(p[0]), int(p[1])) for p in env._apple_pos}
797 12
798 13    # [Single centralized BFS: exact distance + first move to every reachable cell]
799 14    distances, first_moves = {}, {}
800 15    q = deque([(r, c, 0, None)])
801 16    visited = {(r, c)}
802 17    while q:
803 18        cr, cc, d, fm = q.popleft()
804 19        distances[(cr, cc)] = d
805 20        first_moves[(cr, cc)] = fm
806 21        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
807 22            nr, nc = cr + dr, cc + dc
808 23            if 0 <= nr < env.height and 0 <= nc < env.width and not env.walls[nr, nc]:
809 24                if (nr, nc) not in visited:
810 25                    visited.add((nr, nc))
811 26                    q.append((nr, nc, d + 1, fm if fm is not None else (dr, dc)))
812 27
813 28    # [Both alive AND respawning apples in MY zone, with their respawn timer]
814 29    my_zone_spawns = []
815 30    for i in range(env.n_apples):
816 31        pr, pc = int(env._apple_pos[i][0]), int(env._apple_pos[i][1])
817 32        if zones.get((pr, pc)) == agent_id and (pr, pc) in distances:
818 33            my_zone_spawns.append((pr, pc, int(env.apple_timer[i]), distances[(pr, pc)]))
819 34
820 35    # [SPATIOTEMPORAL PRIORITY: earliest collection time first.
821 36    # score = max(walk_distance, respawn_timer); ties broken by sooner timer]
822 37    my_zone_spawns.sort(key=lambda x: (max(x[3], x[2]), x[2], x[3]))
823 38
824 39    for pr, pc, timer, dist in my_zone_spawns:
825 40        if timer == 0: # [Alive apple: walk straight to it]
826 41            if dist == 0: return 7
827 42            fm = first_moves[(pr, pc)]
828 43            return direction_to_action(fm[0], fm[1], orient)
829 44        else: # [Dead apple: camp on safe adjacent cell]
830 45            # ... pick nearest neighbor (nr,nc) that is NOT in spawn_points,
831 46            # ... navigate there and STAND while waiting for respawn
832 47            # ...
833 48    # [Global poaching fallback if zone is empty: nearest alive apple anywhere]
834 49    ...
835 50
836 51

```

838 C.5 Cross-condition observations

839 Several patterns are visible across these four listings (and confirmed by inspecting the remaining 8 runs not shown):

- 841 • *Role assignment encodes the welfare objective.* Static $\text{agent_id} < \tau$ (Listing 5) maximizes U but harms $\min_i R_i$. Time-rotated $(\text{agent_id} + \text{step}/T) \% n$ (Listings 6, 7) maximizes $\min_i R_i$ at $\leq 1\%$ efficiency cost (Gemini).

- 844 • *Convergent rediscovery across LLMs.* Gemini and Sonnet, on independent runs, both arrive at
845 the `(id + phase) % n` duty-rotation idiom with similar phase lengths ($T \approx 50$ steps) under
846 maximin, and both omit it under efficiency.
- 847 • *Game structure dictates strategy class.* Cleanup policies are dominated by role-assignment logic
848 (cleaner vs. gatherer); Gathering policies are dominated by territory partitioning and respawn-
849 timer reasoning, with no role differentiation at all. The researcher does not need to be told which
850 class of strategy to write.
- 851 • *Earliest-collection-time as a unifying score.* The Gathering policy’s `max(walk, timer)` key (List-
852 ing 8) is structurally identical across the 4 Gathering runs (both Gemini and Sonnet), differing only
853 in how the Voronoi diagram is computed, e.g. by Manhattan approximation, fully BFS-based, or
854 cached helper.